

The H_3 Reflection Group

Icosahedral construction, spectral formula, and verification

Computational verification note

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Outcome. Explicit rotation matrices verify the proposed A_4 , S_4 , and A_5 angle formulas in 45 multivariate coefficients each. The 120 matrices of $W(H_3)$ were reconstructed from the icosahedron; all 139 coefficients through total degree ten agree with the signed class-spectrum formula, and the one-variable series agrees with $((1 - t^2)(1 - t^6)(1 - t^{10}))^{-1}$.

1 What is being tested

For a finite matrix group $G \subset GL(V)$, define

$$\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a g)^{-1}. \quad (1)$$

If g has eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\det(I - tg)^{-1} = \sum_{d \geq 0} \chi_{\text{Sym}^d V}(g) t^d.$$

Thus the coefficient at $t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}$ is

$$\frac{1}{|G|} \sum_{g \in G} \prod_a h_{\tau_a}(\lambda(g)) = \dim \left(\bigotimes_a \text{Sym}^{\tau_a} V \right)^G. \quad (2)$$

The last equality is the trace of the Reynolds projector. The test below constructs the matrices in (2) without using the proposed class data, then compares against a separate weighted spectral computation.

2 Floating-point reconstruction diagnostic for $W(H_3)$

Let $\varphi = (1 + \sqrt{5})/2$. Use the twelve vertices

$$\Omega = \{(0, \pm 1, \pm \varphi), (\pm 1, \pm \varphi, 0), (\pm \varphi, 0, \pm 1)\}. \quad (3)$$

Choose an ordered, linearly independent base triple B of vertices. For every ordered target triple T having the same Gram matrix, form the unique linear map $R = TB^{-1}$. Retain R when

$$R^\top R = I, \quad \det R = 1, \quad R\Omega = \Omega.$$

This produces 60 distinct rotations. Adjoining their negatives produces 120 orthogonal transformations. This is the full symmetry group of the icosahedron, which is the real reflection group $W(H_3)$.

This “vertex-frame reconstruction” is useful when no computer algebra class table is available, but the present implementation uses floating-point coordinates and therefore supplies a diagnostic rather than an exact group certificate. The implementation reports its equality tolerance, the minimum separation between reconstructed matrices, the maximum closure-matching distance, and the condition number of the chosen base triple; it then tests every one of the 120^2 products against the reconstructed set. No closure claim here is described as exact.

The regenerated diagnostic reports equality tolerance 5×10^{-10} , minimum separation 1.6625 between distinct reconstructed matrices, maximum closure-matching distance 3.77×10^{-16} , base-triple condition number 1.852, and maximum orthogonality residual 4.16×10^{-16} .

3 Polyhedral rotation groups

For a three-dimensional rotation through angle θ , define

$$\Phi_\theta(\mathbf{t}) = \prod_{a \geq 1} ((1 - t_a)(1 - 2 \cos \theta t_a + t_a^2))^{-1}.$$

The spectrum is $(1, e^{i\theta}, e^{-i\theta})$, so direct substitution of the rotation counts gives

$$\begin{aligned} \mathcal{S}_{A_4, \text{rot}} &= \frac{1}{12}(\Phi_0 + 3\Phi_\pi + 8\Phi_{2\pi/3}), \\ \mathcal{S}_{S_4, \text{rot}} &= \frac{1}{24}(\Phi_0 + 6\Phi_{\pi/2} + 8\Phi_{2\pi/3} + 9\Phi_\pi), \\ \mathcal{S}_{A_5, \text{rot}} &= \frac{1}{60}(\Phi_0 + 15\Phi_\pi + 20\Phi_{2\pi/3} + 12\Phi_{2\pi/5} + 12\Phi_{4\pi/5}). \end{aligned}$$

The direct route independently constructs the 12 tetrahedral matrices as the determinant-one signed permutation matrices preserving a fixed tetrahedron, the 24 octahedral matrices as all determinant-one signed permutation matrices, and the 60 icosahedral matrices by the vertex-frame construction above. For each group, all 45 coefficients indexed by partitions through total degree seven agree with the displayed angle formula.

group	order	coefficient checks	result
A_4 tetrahedral rotations	12	45	PASS
S_4 octahedral rotations	24	45	PASS
A_5 icosahedral rotations	60	45	PASS

4 The explicit H_3 spectral formula

Write

$$A_z(\mathbf{t}) = \prod_{a \geq 1} (1 - z t_a)^{-1}, \quad \omega = e^{2\pi i/3}, \quad \xi = e^{2\pi i/5}.$$

The orientation-preserving subgroup has the following spectra and class sizes:

size	eigenvalues	geometric type
1	$(1, 1, 1)$	identity
15	$(1, -1, -1)$	half turns
20	$(1, \omega, \omega^2)$	third turns
12	$(1, \xi, \xi^{-1})$	first fifth-turn class
12	$(1, \xi^2, \xi^{-2})$	second fifth-turn class

Multiplication by central inversion negates all three eigenvalues and supplies the other five classes. Grouping (1) by these ten spectra gives

$$\begin{aligned} \mathcal{S}_{H_3} = \frac{1}{120} & \left[A_1^3 + A_{-1}^3 + 15(A_1 A_{-1}^2 + A_{-1} A_1^2) \right. \\ & + 20(A_1 A_\omega A_{\omega^2} + A_{-1} A_{-\omega} A_{-\omega^2}) \\ & \left. + 12 \sum_{j=1}^2 (A_1 A_{\xi^j} A_{\xi^{-j}} + A_{-1} A_{-\xi^j} A_{-\xi^{-j}}) \right]. \end{aligned} \quad (4)$$

Equation (4) is exact in all multidegrees. It is more informative than the ordinary invariant degrees because mixed coefficients depend on the entire class spectrum.

4.1 Ordinary Molien specialization

Setting only $t_1 = t$ nonzero gives

$$\mathcal{S}_{H_3}(t, 0, \dots) = \frac{1}{(1-t^2)(1-t^6)(1-t^{10})}. \quad (5)$$

The right side has degree- d coefficient equal to the number of nonnegative solutions of $2a + 6b + 10c = d$. This elementary count is used as a third route, independent of both explicit matrix spectra and the multivariate class calculation.

5 Verification protocol

1. Reconstruct the 60 rotations from (3), adjoin negatives, and test order, orthogonality, signed determinants, and closure.
2. For every partition τ through total degree ten, compute the direct matrix average in (2) over all 120 matrices.
3. Compute the same coefficient using only the ten weighted spectra in (4). The explicit matrices are not passed to this route.
4. For every $0 \leq d \leq 10$, compare the direct one-variable average with the solution count for $2a + 6b + 10c = d$.
5. Construct the six-dimensional $\text{Sym}^2(V)$ matrices and average them to a Reynolds projector.

6 Results

Check	observed	status
number of reconstructed matrices	120	PASS
closure products tested	14,400	PASS
maximum $\ g^T g - I\ _F$	4.16×10^{-16}	PASS
signed determinant condition	all matrices	PASS
partitions through degree 10	139 coefficients	PASS
ordinary Molien degrees 0 through 10	11 coefficients	PASS

Some matched multivariate coefficients are

τ	(1)	(2)	(3)	(4)	(2, 1)	(2, 2)	(3, 2)
dimension	0	1	0	1	0	2	0

The coefficients of (5) for $d = 0, \dots, 10$ are

$$1, 0, 1, 0, 1, 0, 2, 0, 2, 0, 3,$$

and agree termwise with direct matrix averaging.

For $\tau = (2)$, the target $\text{Sym}^2(V)$ has dimension six. The explicit Reynolds projector has

$$\text{rank } P = 1, \quad \text{tr } P = 0.9999999999999998, \quad \|P^2 - P\|_F = 1.26 \times 10^{-16}.$$

The invariant is the expected quadratic form.

7 Reproducibility and limitations

Run from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.h3_reflection_verification
```

The companion marimo notebook exposes the maximum tested degree and displays all coefficient rows.

The group reconstruction uses double-precision values of φ , not an exact quadratic-number field. This does not affect the theoretical proof of (4), but it makes the group diagnostics numerical. To control this, vertices and matrices are canonicalized to nine decimal places, every product is tested, and all averaged characters must lie within 5×10^{-7} of an integer. The actual orthogonality and projector residuals are around 10^{-16} .

The finite checks are diagnostic evidence; the all-degree validity of (4) comes from the master identity and the complete class spectra. The computation specifically verifies that the class weights, signs, fifth-root classes, and matrix convention have been transcribed consistently.